PhET Simulator: _____ Energy Skate Park!

Name:

Date: _____

Period: _____

1) Type in the web address: https://phet.colorado.edu/en/simulation/legacy/energy-skate-park and click the "**Play**" button.

2) Choose which skater you'd like to use (third option down on right) then "**x**" out of option window.

3) Click on "**Bar Graph**". Watch as the skater moves across the track and how it changes the levels of Potential and Kinetic Energy. (You can slow or pause the skater down at the bottom of the simulator window to help you observe).

   a) When is the **gravitational potential energy** the highest? _____ The lowest? _____

   b) When is the **kinetic energy** the highest? _____ The lowest? _____

   c) When are the kinetic and gravitational potential energy levels the ***same***? _____

   d) Can you think of a scenario when the kinetic and gravitational potential energy could both be ***zero***? **Describe** or **draw** how this could be possible below:

4) Notice the 3 blue circles, one at each end of the track and one in the middle. Click on one circle and hold down. Move the circle to change the track's shape to however you'd like and observe how the track responds. When you feel comfortable manipulating the track, click "**Reset**" and Select "**Yes**".

5) Move the circles to make the **right** side of the track **LONGER** than the **left** side of the track. Click on "**bar graph**".

   a) Bring the skater up to various heights on the **left** side of the track (at the top, half way up, etc.) and then observe what happens when you let them go. Do this until you can predict to what height the skater will reach on the right side of the track. Pick a height of release and sketch the path the skater will take when you release them:

b) Use the **Law of Conservation of Energy** to explain how you can predict the height that the skater will reach on the ***right*** side depending on where you release them on the ***left*** side:

_____

_____

_____.

6) Now make the **right** side of the track **SHORTER** than the **left** side of the track. What do you think will happen if you release the skater from the top of the left side of the track? Use the relationship between gravitational potential energy and kinetic energy to explain your thinking and then test to see if your prediction was correct:

_____

_____

_____.

7) Notice the yellow window labeled "**Tracks**" in the top left corner. This feature allows you to build your own track in the simulator window. All you have to do is select and drag the new piece of track to the existing track and line up the blue circles for where you want them to join. Then, you can manipulate the track to create hills, loops, drops, etc! Practice getting familiar with creating new shapes and when you're ready, click "**Reset**" and select "**Yes**".

8) Now you are going to create your own skatepark! Your mission is to create a track complete with hills, drops and multiple loops that your skater can complete from start to finish without getting stuck OR flying off! Carefully consider the ***relationship between potential and kinetic energy*** and then sketch your plan below!

9) Once your sketch is complete, test to see if your skater is able to successfully complete the track from start to finish!

a) Was your skater successful on your first attempt? _____

b) If no, indicate on your sketch where your skater became stuck or flew off. Then, edit your plan and try again! Indicate any updates you make using a **different color** or **dotted line**- create a key so we can clearly see any changes you've made!

10) On your sketch above, clearly label the points of **greatest PE and KE** and **least PE and KE**.

***Use the options on the right to see if your track can work in outer space! On the Moon? On Jupiter? ***